ripple-effect performances and the GradSim values, validating that gradient-based knowledge storage similarity is a reliable indicator of ripple effects. Additionally, we observe the emergence of two distinct clusters in the figure. This clustering likely occurs because the data points can be categorized into successful and unsuccessful edits. Successful edits result in a significant improvement on performance, placing them in the upper successful cluster, while unsuccessful edits tends to remain in the unsuccessful cluster.

3.4 Counter-Intuitive Failure Cases

Negation Negation is one of the most straightforward ripple effects where the model is expected to answer a negated query after an editing is applied. For example, after editing the nationality of *Leonardo DiCaprio* as *Syrian*, the model should be able to avoid *Syria* given a negated query like "*Leonardo DiCaprio is* ***not*** *a citizen of*". However, both smaller-sized LMs like GPT-2 and larger-sized LMs like LLaMA still answers "Syria" to this query and simply ignore the negation inside the sentence. In Figure 3, we first visualize both the values and gains of model likelihoods for the original and negated facts. The results demonstrate a strong positive (almost linear) correlation, indicating a severe problem of negation failures. In terms of GradSim values, we find that the gradient similarities between the original and negated facts are very high, suggesting that the original and negated facts are entangled in similar knowledge storage locations.

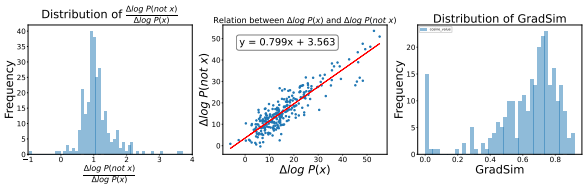


Figure 3: Correlation between original and negated facts on likelihood change, and the distributions of GradSim and likelihood ratios between original and negated facts.

Over-Ripple Errors The over-ripple problem refers to the situation where, after a knowledge edit, the LM only memorizes the edited target itself and continues to provide this target as the answer even when asked about other knowledge that is related. For example, after editing the nationality of *Leonardo DiCaprio* as *Syrian*, the model will still answer *Syria* even when asked about the primary language Leonardo is speaking (the correct answer should be *Arabic*). In Figure 4, we first visualize GradSim distributions on (q_y, a'_x) and (q_y, a'_y) respectively, and we can observe that the edited target a'_x (e.g., *Syria*) has a much higher gradient similarity compared to the correct answer a'_y (e.g., *Arabic*). This explains the similar performance of answering both the correct and incorrect answers and indicates the occurrence of over-ripple errors.

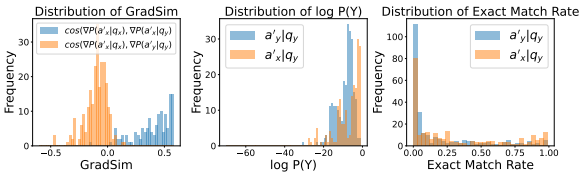


Figure 4: The distributions of GradSim values and ripple effect performances.

Cross-Lingual Transfer The problem of cross-lingual transfer is defined as the ability to edit a piece of knowledge in one language and have the model still provide the correct answer when asked a question in another language. We study the role of GradSim by visualizing the distribution of GradSim values and the ripple effect performance across different languages. We employ Baichuan (Yang et al., 2023), the state-of-the-art bilingual model for Chinese and English. As shown in Figure 5, while the performance on the target language remains low, the GradSim values are also very low, primarily distributed near zero. GradSim works as a reliable indicator in this special ripple-effect case for cross-lingual transfer.

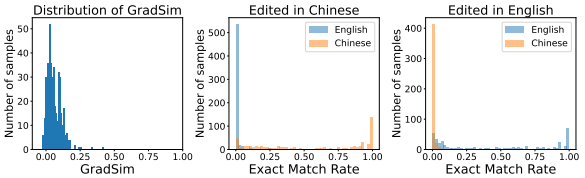


Figure 5: GradSim and performance distributions when editing on one language and testing on another.

4 Conclusion

Through extensive experiments and analysis, we propose GradSim, computed as the cosine similarity between gradients of the original fact and its related knowledge, as a crucial indicator for the effectiveness of the ripple effects. The positive correlation observed between GradSim and ripple effect performance across various LMs, KE methods, and evaluation metrics underscores its reliability. Additionally, our exploration of failure cases further confirms that low GradSim values are indicative of ripple effect failures.

Limitations

The first notable limitation is that, although a strong relationship between GradSim and the performance of ripple effects has been demonstrated, our research remains at the level of exploring correlations between these two factors and has not yet established a causal relationship. While it is always challenging to determine causality, it would still be extremely interesting and exciting to explore the dominant contributing factors to the complex distribution of knowledge storage in the pre-training phase of LMs. The second important limitation is that, while this paper identifies an indicator, we did not provide practical solutions for improving ripple effect performance by leveraging this indicator. However, we believe that the insights provided in this short paper will significantly enable the development of practical and effective methods in future research.

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A Knoweledge Editing Methods

Current knowledge editing methods can be categorized into several distinct approaches, including fine-tuning-based methods, locate-then-edit methods, and meta-learning methods. In this paper, we employ a range of knowledge editing techniques to validate our theorem. A brief introduction to each of these methods will be provided here.

ROME: Rank-One Model Editing (ROME) conceptualizes an MLP module as a key-value store. In this framework, the key represents an encoded subject, while the value represents the knowledge associated with that subject. The MLP retrieves the corresponding value by accessing the key. ROME modifies the MLP weights using a rank-one adjustment to directly insert new key-value pairs.

MEMIT: Mass Editing Memory in a Transformer (MEMIT) builds upon ROME, and is designed to handle large-scale edits by inserting multiple memory entries simultaneously through modifications to the MLP weights across several key layers. MEMIT employs causal tracing to identify a set of mediating MLP layers that store and recall memories related to a specific subject. For a set of new memories, an update (Δ) is computed and propagated across all the identified mediating MLP layers, ensuring that by the final layer, the output captures and reflects all the newly inserted memories.

MEND: Model Editor Networks with Gradient Decomposition (MEND) consist of small auxiliary networks designed to make quick, localized edits to a pre-trained model's behavior using a single input-output pair. MEND achieves this by learning how to transform the gradient generated through standard fine-tuning, employing a low-rank decomposition of the gradient to make the transformation more computationally feasible. MEND can be trained efficiently on a single GPU in less than a day, even for models with over 10 billion parameters. Once trained, MEND allows for rapid application of new edits to the pre-trained model.

B Additional Experiments

B.1 Experiments with various knowledge experiments

To support our theorem, we also did experiments with more knowledge editing methods, including Fine-Tuning(FT) and MEND. In the basic FT process, we use Adam and early stopping to minimize the loss of new edits on full parameters. In the